Answer Key — Electromagnets

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) electromagnet** — a coil carrying current that acts as a magnet is an electromagnet.

Q2 **(c) increase the current and number of turns** — more current and more turns (and an iron core) make it stronger.

Q3 **(c) two poles, North and South** — like a bar magnet, it has both a North and a South pole.

Q4 **(a) True** — more cells → more current → stronger magnet → more clips.

Q5 **(a) Both true, R explains A** — the iron core strengthens the magnet, which is why it is used.

Q6 **direction** — reversing current direction reverses the poles.

Q7 **lifting** — lifting electromagnets are used in scrap yards.

Short answer

Q8 A current-carrying coil that acts as a magnet. Stronger by increasing the current (more cells), increasing the number of turns, or inserting an iron core.

Q9 It can be switched on and off, and its strength and polarity can be changed (by current/turns or current direction).

Q10 They fall off, because the magnetic field disappears when the current stops, so it is no longer a magnet.

Long / application

Q11 Current through the coil (with iron nail core) creates a magnetic field, making the nail a magnet that attracts clips; opening the circuit stops the current, the field vanishes, and clips fall. If the cell weakens, the current drops, so the magnetic field is too weak to hold the clips.

Q12 Yes, the coil alone still produces a magnetic field and deflects the compass, but the deflection is **less** than with the iron nail, because the iron core greatly strengthens the magnet.

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